AI 时代商业工程 · 课题研究系列
BUSINESS ENGINEERING IN THE AI ERA · RESEARCH STUDY SERIES

No. 06 / 12

课题研究报告 · RESEARCH STUDY

上下文工程：为模型所读之物建立预算、底座与治理

Context Engineering: Budget, Substrate and Governance for What Models Read

关于窗口预算、图谱底座与读取治理的课题研究

A research study on the window budget, the graph substrate and the governance of the read

研究底本 SOURCE
Cuofano, G. Context Engineering — The Graph Engineer's Discipline (2026)

版本 EDITION
2026 年 10 月 · October 2026

语言 LANGUAGE
中英双语 · Bilingual Chinese–English

摘要 *Abstract*

语言模型在使用时刻所能获知的一切，都只能经由其上下文窗口进入。本研究以 Cuofano（2026）《Context Engineering》为底本，将这一「唯一之门」命题作为分析起点，考察上下文如何同时决定智能体系统的**能力、成本与治理**。研究提出四个问题：窗口为何应被视为预算而非容器；存储的形态如何决定智能体可提出的问题；治理规则为何只有落在上下文层才具有结构性约束力；以及这一学科应如何被度量与组织。方法上，本研究对底本的三十六章、四十三幅图版、一个端到端复合案例与七个公开案例进行结构化阅读，并将底本自述的四项可证伪命题整理为研究假设。研究得出四项主要发现：其一，注意力随窗口填充而稀释、计算成本随长度平方增长，因此常驻层应当「薄而指向深处」，放置与压缩都是需要显式决策的工程动作；其二，文档、表、向量与图四种存储形态分别回答不同类型的问题，而智能体以「描述」提问，只有附带来源与权限的图谱底座能原生作答；其三，提示词中的规则只是建议，而附着在事实上的来源、读取时的权限、工具的作用域与回写的闸门才构成真正的控制；其四，上下文需要独立的评测套件、读取日志与六项指标。复合案例显示，在模型不变的情况下，输出套件通过率由 71% 升至 89%。本研究最后讨论了对组织与「图工程师」角色的启示、证据层面的局限，并提出未来研究议程。

Everything a language model can know at the moment of use enters through its context window. Taking Cuofano's (2026) *Context Engineering* as its source volume, this study treats that "only door" as its point of departure and examines how context jointly decides the **capability, cost and governance** of agentic systems. It asks four questions: why the window should be treated as a budget rather than a container; how the shape of a store decides which questions an agent can ask; why rules bind structurally only when they live in the context layer; and how the discipline should be measured and organised. Methodologically, the study reads the source's thirty-six chapters, forty-three plates, one end-to-end composite case and seven public cases in a structured way, and recasts the four falsifiable patterns the source states as hypotheses. Four findings emerge. Attention dilutes as the window fills and compute grows with the square of its length, so the standing layer must be thin and point at depth, while placement and compaction are explicit engineering decisions. Document, table, vector and graph stores answer different kinds of question, and agents ask by description, which only a graph substrate carrying provenance and permission answers natively. Rules in the prompt are advisory; provenance on the fact, permission on the read, scope on the tool and a gate on the write are the real controls. And context needs its own suite, read log and meters. In the composite case the output pass rate rises from 71% to 89% with the model unchanged. The study closes with organisational implications, limitations and a research agenda.

关键词 上下文工程；上下文窗口；知识图谱；来源与权限；提示注入；图工程师

KEYWORDS context engineering; context window; knowledge graph; provenance and permission; prompt injection; graph engineer

研究问题 *Research Questions*

RQ1 在注意力稀释与平方成本的约束下，上下文窗口应如何被分配、排布与压缩?

Under attention dilution and quadratic cost, how should the context window be allocated, ordered and compacted?

RQ2	存储的形态如何决定智能体能够提出的问题，图谱底座在其中扮演何种角色?	How does the shape of a store decide which questions an agent can ask, and what role does a graph substrate play?
RQ3	为什么治理规则只有在上下文层才具有结构性约束力，其具体控制点是什么?	Why do governance rules bind structurally only in the context layer, and what are the concrete control points?
RQ4	应以何种评测、指标与组织形态来度量并维系上下文工程这一学科?	What evaluations, meters and organisational forms should measure and sustain context engineering as a discipline?

目录 *Contents*

01 研究背景与问题提出
Background and Problem Statement 5

02 核心概念与理论框架
Core Concepts and Analytical Framework 7

03 研究设计与方法
Research Design and Method 9

04 发现一：窗口是预算，而非容器
Finding 1: The Window Is a Budget, Not a Container 11

05 发现二：存储的形态决定可问之题
Finding 2: The Shape of the Store Decides the Question 14

06 发现三：治理在上下文层成为结构
Finding 3: Governance Becomes Structure in the Context Layer 17

07 案例研究：一个底座的端到端演进
Case Study: One Substrate, End to End 20

08 讨论：启示与风险
Discussion: Implications and Risks 23

09 结论、局限与未来研究方向
Conclusions, Limitations and Research Agenda 25

— 参考文献与研究说明
References & Research Notes 27

第 1 章 · CHAPTER 1

01 研究背景与问题提出

Background and Problem Statement

模型在使用时刻只知道它读到的东西。若窗口是唯一之门，那么能力、成本与治理便都在这扇门前被决定。

中文

At the moment of use a model knows only what it reads. If the window is the only door, then capability, cost and governance are all decided at that door.

ENGLISH

1.1 「唯一之门」命题

The “Only Door” Proposition

底本的出发点是一个看似朴素的事实：模型的权重承载着巨大的压缩先验，但在使用的那一刻，它看不到企业的系统，记不得昨天，也不了解桌上的那份文件。它只有一个孔径，即本轮放在它面前的词元序列。关于企业世界它能知道的一切、能做的一切、可能泄露的一切，都要经过这一孔径。本研究把这一命题称为「唯一之门」，并把它作为全部分析的公理起点。

The source starts from a deceptively simple fact. A model’s weights carry an enormous compressed prior, but at the moment of use it has no view of the firm’s systems, no memory of yesterday and no knowledge of the document on the desk. It has a single aperture: the sequence of tokens placed before it on this turn. Everything it can know about the firm’s world, everything it can do in it and everything it can leak from it passes through that aperture. This study calls the proposition the “only door” and treats it as the axiom from which the analysis proceeds.

由此推出三个判断。**能力**在窗口中决定：有正确事实的模型会作答，没有的则会流畅地猜测。**成本**在窗口中决定：每个词元在每一轮都要付费，而模型能分给任一词元的注意力随窗口填满而下降。**治理**同样在窗口中决定：写在提示词里的规则只是一项请求，而在决定何物进入窗口的那一层被执行的规则，才是结构。模型无法违反一项关于它从未见过之事实的权限。

Three judgements follow. **Capability** is decided in the window: a model with the right fact answers, and one without it guesses fluently. **Cost** is decided there too: every token is paid for on every turn, and the attention the model can give any one token falls as the window fills. And **governance** is decided there: a rule written in the prompt is a request, while a rule enforced in the layer that decides what enters the window is structure. A model cannot violate a permission on a fact it was never shown.

概念界定 DEFINITION

上下文工程 *Context engineering*

决定模型读什么的学科：什么常驻于每一轮的窗口、什么在轮次之间持久保存、什么按需获取、什么被写回，以及什么被有意排除。其对应的角色是「图工程师」，负责构建使这些决定可被治理的底座。

The discipline of deciding what the model reads: what stands in the window every turn, what persists between turns, what is fetched on demand, what is written back and what is deliberately left out. Its corresponding role is the graph engineer, who builds the substrate that makes those decisions governable.

1.2 为何是现在：六股汇聚的力量
Why Now: Six Converging Forces

底本认为，这一学科在 2023 年为时尚早，到 2030 年将被视为理所当然，而在当前阶段具有决定性，因为六股相互叠加的技术力量同时指向窗口这一孔径（图 1-1）。

The source argues that this discipline would have been premature in 2023 and will be assumed by 2030, but is decisive now, because six compounding technical forces converge on the window (Figure 1-1).

图 1-1 六股力量汇聚于窗口
Figure 1-1 *Six forces converging on the window*

前沿趋同 Frontier converged	模型差距以周计，环绕模型之物成为产品	Models weeks apart; what surrounds the model becomes the product
推理时增益 Inference-time gains	推理、工具调用与检索都是窗口操作	Reasoning, tool calls and retrieval are all window operations
智能体行动 Agents act	读到的即是要做的，窗口成为治理面	What is read is what is done; the window becomes a governance surface
单任务词元上升 Tokens per task rise	单价下降，单位结果的账单反而上升	Unit price falls while the bill per outcome rises
第一桶油枯竭 First oil runs out	企业工作中产生的「残留物」成为耐久材料	The residue of real work becomes the durable material
读者改变 The reader changed	机器按描述提问，答案引擎偏好结构化来源	Machines ask by description; answer engines favour structured sources

来源 Source: Cuofano (2026), “Why Now”.

>1M
前沿窗口词元数
frontier window tokens

~1000×
五年间窗口增长
window growth in five years

2025
「上下文工程」公开命名
term named in public

来源 Source: Cuofano (2026), Chapter IV and Appendix B, as read in Q3 2026.

1.3 问题的提出
Framing the Problem

底本指出，多数企业 AI 项目把层次放反了：治理放进提示词而沦为建议，上下文放进检索盒子而沦为管道。本研究的核心问题因此是：若模型是租来的、底座是自有的，企业应如何设计模型所读之物，使之在能力上充分、成本上可控、治理上可执行?

The source observes that most enterprise AI programmes have the layers the wrong way round: governance in the prompt, where it is advisory, and context in a retrieval box, where it is plumbing. This study’s core question is therefore: if the model is rented and the substrate owned, how should a firm design what its models read so that it is sufficient for capability, controllable in cost and enforceable in governance?

第 2 章 · CHAPTER 2

02 核心概念与理论框架

Core Concepts and Analytical Framework

本章建立全文的分析框架：四个平面与「提示词不是政策」两项命题，以及由此导出、构成闭环的五个动作。

中文

This chapter sets out the analytical framework: the four planes and the proposition that the prompt is not the policy, together with the five moves they imply, arranged as a loop.

ENGLISH

2.1 四个平面
The Four Planes

底本把生产中的 AI 系统划分为四个平面，并认为多数故障源于一个平面在做另一个平面的工作。应用平面负责「做什么」；执行平面即完成工作的循环，负责「怎么做」；上下文平面是模型所读的一切，负责「用什么做」；治理平面是规则，负责「是否做、以何条件做」。平面之间存在向下依赖：应用交付不了执行平面产出不了的结果，执行平面产出不了上下文平面未提供事实支撑的结果。

The source divides a production AI system into four planes and argues that most failures are one plane doing another plane's job. The application plane owns the what; the execution plane, the loop that does the work, owns the how; the context plane, everything the model reads, owns the with-what; and the governance plane, the rules, owns the whether and on what terms. The planes depend downward: the application cannot deliver an outcome the execution plane cannot produce, and the execution plane cannot produce one for which the context plane did not supply the facts.

图 2-1 四个平面及其职责
Figure 2-1 *The four planes and what each owns*

应用平面 Application	界面、端点与交付的结果；客户唯一可见的平面	Interface, endpoint and delivered outcome; the only plane customers see
执行平面 Execution	模型、工具、编排、重试与裁判组成的循环	The loop of model, tools, orchestration, retries and referee
上下文平面 Context	窗口、记忆、检索、记录与底座；通往世界的唯一之门	Window, memory, retrieval, record and substrate; the only door to the world
治理平面 Governance	谁可对何物做何事；自身没有执行机制	Who may do what to what; it has no mechanism of its own

来源 Source: Cuofano (2026), Chapter I.

第二种关系颠覆了常规设计：治理平面没有自己的执行机制。能在物理上执行规则的只有执行平面（智能体无法调用的工具）和上下文平面（智能体无法看到的事实）。后者更强，因为被扣留的能力仍可被发现，被扣留的事实则根本不存在。窗口自身的物理约束，即平方成本与注意力稀释，将在第四章展开。

The second relationship inverts the usual design: the governance plane has no mechanism of its own. Enforcement can physically happen only in the execution plane, as a tool the agent cannot call, and the context plane, as a fact it cannot see. The latter is stronger, because a withheld capability is still discoverable while a withheld fact is simply absent. The window's own physics, quadratic cost and attention dilution, is taken up in Chapter 4.

核心洞见 KEY INSIGHT

治理平面即上下文平面 *The governance plane is the context plane*

治理是意图的平面，上下文是大部分意图变成结构的地方。一条不在窗口里起作用的规则不是规则；一项不附着在事实上的权限不会被执行。

Governance is a plane of intent; context is where most of that intent becomes structure. A rule that does not act on the window is not a rule; a permission not carried on the fact is not enforced.

2.2 提示词不是政策
The Prompt Is Not the Policy

企业 AI 最常见的治理架构是提示词顶部的一段话。对一个把全部输入视为文本的系统而言，这只是一项请求：模型通常会遵守，但在对抗压力下会落败。把规则下移一层，让薪资事实带着权限存入底座、检索层对不满足权限的身份一无所返，便既无需指令，也无指令可被覆盖。

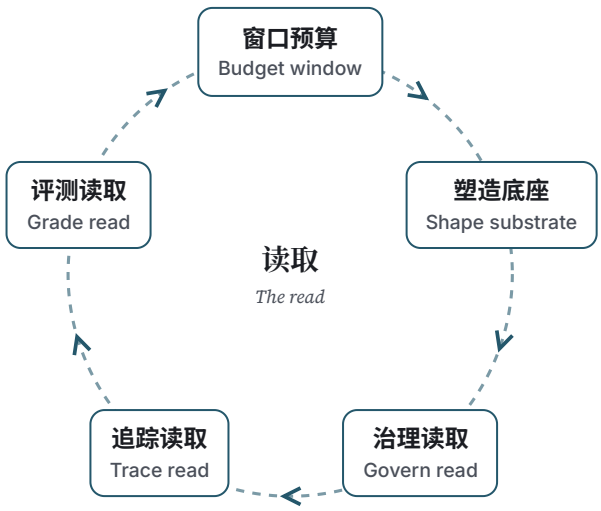
The commonest governance architecture in enterprise AI is a paragraph at the top of the prompt. To a system that treats all input as text it is only a request: the model usually complies and, under adversarial pressure, loses. Move the rule one plane down, so salary facts sit in the substrate with a permission and retrieval returns nothing to an identity that fails it, and no instruction is needed and none can be overridden.

2.3 五个动作：本研究的分析框架
The Five Moves: This Study's Framework

底本将学科压缩为五个动作。本研究把它们组织为闭环（图 2-2）：预算约束窗口，底座决定可问之题，治理决定可见之物，追踪留下证据，评测再修订预算与底座。三个发现章分别对应前三者，追踪与评测贯穿案例与讨论。

The source compresses the discipline into five moves, which this study arranges as a loop (Figure 2-2): the budget constrains the window, the substrate decides what can be asked, governance decides what can be seen, tracing leaves evidence and grading revises budget and substrate. The three findings chapters take the first three; tracing and grading run through the cases and discussion.

图 2-2 五个动作构成的闭环
Figure 2-2 *The five moves as a closed loop*



来源 Source: Cuofano (2026), Part Six; loop arrangement is this study's.

第 3 章 · CHAPTER 3

03 研究设计与方法

Research Design and Method

本研究采用对单一底本的结构化分析阅读，以章节、图版与案例为分析单位，并把底本自带的可证伪命题整理为假设。

中文

The study uses a structured analytical reading of one source volume, takes chapters, plates and cases as units of analysis, and recasts the source's own falsifiable patterns as hypotheses.

ENGLISH

3.1 资料与分析单位

Material and Units of Analysis

分析材料为底本的全部正文：三十六章、四十三幅图版、五个附录与注释。分析单位分为三类：一是**机制**，即底本陈述的因果链条，如注意力稀释或权限在检索时的执行；二是**案例**，包括附录 C 的现场服务智能体复合案例和附录 D 的七个公开案例；三是**读数**，即底本标注为 2026 年第三季度的时点测量。本研究对机制做比较归类，对案例做前后对照，对读数只作引用、不作外推。

The material is the whole of the source: thirty-six chapters, forty-three plates, five appendices and the notes. There are three kinds of unit of analysis. **Mechanisms** are the causal chains the source states, such as attention dilution or permission enforced at retrieval. **Cases** are the field-service composite in Appendix C and the seven public cases in Appendix D. **Readings** are point-in-time measurements the source dates to Q3 2026. Mechanisms are compared and classified, cases are read before and after, and readings are cited but never extrapolated.

36	43	7	4
章节 chapters	**图版** plates	**公开案例** public cases	**可证伪命题** falsifiable patterns

来源 Source: Cuofano (2026), contents, Appendices A and D.

3.2 研究假设

Hypotheses

底本在附录 A 中声明，其承重性模式均附带「证伪条件」，即一旦观察到特定反例便应撤回。本研究据此将四项模式整理为可检验的假设，并为每项列出底本给出的检验工具。这一做法使本研究的发现不仅是对底本观点的归纳，也明确指出了哪些观察会推翻它们。

In Appendix A the source states that its load-bearing patterns carry falsifiers: each is retired by a specific contrary observation. This study recasts four of them as testable hypotheses and lists the instrument the source names for testing each. This makes the findings more than a summary of the source's views; it also states plainly which observations would overturn them.

表 3-1 研究假设及其检验方式
Table 3-1 *Hypotheses and how they are tested*

编号 No.	假设 Hypothesis	检验工具 Test instrument
H1 H1	放置位置显著影响在场事实是否被使用 Placement measurably changes whether a present fact is used	放置测试 Placement tests
H2 H2	事实携带来源比提示词防御更能减少成功注入 Provenance on the fact reduces successful injection versus prompt-level defences	投毒测试 Poisoning tests
H3 H3	同等词元预算下，薄常驻层加按需检索优于厚常驻层 At equal token budget, a thin standing layer with on-demand retrieval beats a thick one	输出套件与读取日志 Output suite and read log
H4 H4	基于成形存储的描述式检索能回答相似度检索失败的关系型问题 Retrieval by description over a shaped store answers relational questions similarity retrieval fails	检索召回率 Retrieval recall

来源 Source: Cuofano (2026), Appendix A; numbering is this study's.

方法说明 METHOD NOTE

证据分层 *Evidence tiers*

底本区分三类陈述：经公开文献与报道核实的案例、以复合形式呈现的工作示例（无真实企业，但每一步真实），以及作为方法提出、附带证伪条件的模式。本研究在引用时保持这一区分，复合案例的数字仅作为机制示例。

The source distinguishes three kinds of statement: cases verified from public documentation and reporting, worked composites (no real firm, every step real), and patterns stated as method with falsifiers. This study keeps the distinction when citing; the composite's numbers serve only as an instance of mechanism.

本研究的局限在方法层面即已确定：它依赖单一底本，未进行独立实验，复合案例的改善幅度不能被视为普遍效应。这些局限将在第九章集中讨论。

The study's limits are fixed at the level of method: it relies on one source volume, runs no independent experiments, and the improvements in the composite case cannot be read as general effects. Chapter 9 discusses these limits in full.

第 4 章 · CHAPTER 4

04 发现一：窗口是预算，而非容器

Finding 1: The Window Is a Budget, Not a Container

把窗口当作水桶的工程师，是在为线性的稀释支付平方级的成本。常驻层、放置、压缩与预算四个杠杆决定了窗口的质量。

中文

The engineer who treats the window as a bucket pays quadratic money for linear dilution. Four levers decide the window's quality: the standing layer, placement, compaction and the budget.

ENGLISH

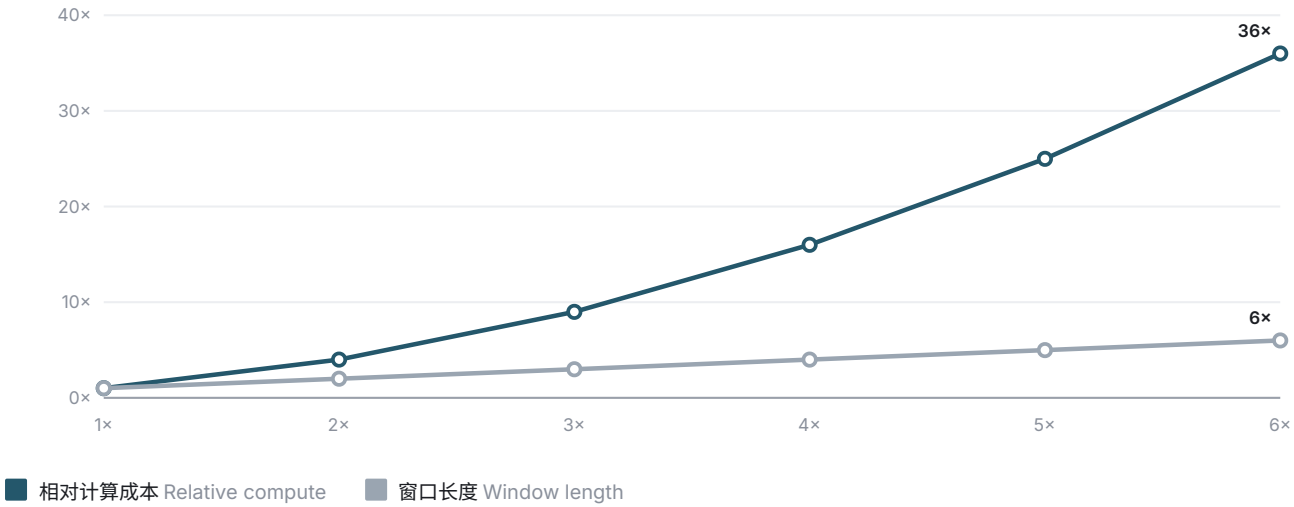
4.1 平方成本与注意力稀释

Quadratic Cost and Attention Dilution

底本由窗口物理学导出三条实践定律。第一，过满与过空同样可靠地导致退化：把一切可能相关的内容塞进窗口不是谨慎，而是稀释。第二，放置是一项决定而非偶然：位于首尾的内容会被读到，位于中间的较少被读到。第三，每个词元在每一轮都有价格，因此常驻层是多数智能体系统中最大、也最被忽视的经常性成本。

From the physics of the window the source derives three practical laws. First, overfilled degrades as reliably as underfilled starves: adding everything that might be relevant is not caution but dilution. Second, placement is a decision, not an accident: what sits first and last is read, and what sits in the middle is read less. Third, every token has a price on every turn, so the standing layer is the single largest and most neglected recurring cost in most agentic systems.

图 4-1 窗口长度与相对计算成本（按平方关系推算）
Figure 4-1 *Window length and relative compute cost (derived from quadratic scaling)*



示意 Illustrative: values derived from the source's statement that compute grows with the square of window length (Cuofano 2026, Ch. II); not measured data.

4.2 常驻层：地图，而非手册

The Standing Layer: Map, Not Manual

常驻层是系统的持久指令：智能体是谁、可以做什么、应如何行事、到哪里去找。人们倾向于把它写成手册，因为手册让人感到安全。底本认为恰恰相反：常驻层中的手册在每一轮都被付费、在每一轮都稀释注意力，而且越长越

The standing layer is the system's persistent instruction: who the agent is, what it may do, how it should behave and where to look. The temptation is to write it as a manual, because a manual feels safe. The source argues the opposite: a manual in the standing

不可靠。业界收敛出的规则是「地图而非手册」：常驻层只说明深层内容在哪里、如何到达，深层内容则在需要时获取。

- **模型本来就知道吗？** 若是，则写入即为稀释。
- **它适用于每一轮吗？** 若否，它属于检索。
- **它是一条政策吗？** 若是，它属于上下文层的权限，而非提示词文字。

layer is paid for on every turn, dilutes attention on every turn and is read least reliably where it is longest. The rule the field converged on is the map, not the manual: the standing layer says where depth lives and how to reach it, and the depth is fetched when needed.

- **Would the model already know this?** Then it is dilution.
- **Does it apply to every turn?** If not, it belongs in retrieval.
- **Is it a policy?** Then it belongs in the context layer's permissions, not in the prose.

4.3 放置与压缩
Placement and Compaction

放置被视为逐项决定。常驻层位于最前，为一切定框；当前问题与最关键的事实位于最后；较长的检索材料位于中间，凡其中必须据以行动的内容都在末尾重述；工具结果按到达顺序放置并随时间修剪。每类条目只有四种去向：常驻、检索、工具获取或省略。底本称放置是学科中最便宜却最少被使用的杠杆：同样的事实，换一种顺序，就是另一个系统。

Placement is treated as a per-item decision. The standing layer sits first, framing everything; the current question and the most decision-critical facts sit last; long retrieved material sits in between, with anything that must be acted on restated near the end; tool results are placed as they arrive and trimmed as they age. Each class of item has one of four outcomes: standing, retrieved, tool-fetched or omitted. The source calls placement the cheapest and least used lever: the same facts in a different order are a different system.

长时运行终会超出窗口，压缩随之成为必然。每次压缩都是一种损失，问题在于哪些损失是安全的。丢弃推理而保留决定，通常安全；丢弃产生该决定的约束则不安全，因为下一轮会在缺少约束的情况下重新推出决定。工程上的回答是结构化：决定、约束与未决问题被压缩为有类型的条目而非散文；不可逆操作在发生时即被记录在窗口之外；压缩本身也接受评测。

Long runs outgrow the window, so compaction becomes unavoidable. Every compaction is a loss, and the question is which losses are safe. Dropping the reasoning while keeping the decision is usually safe; dropping the constraint that produced the decision is not, because the next turn will re-derive the decision without it. The engineering answer is structural: decisions, constraints and open questions are compacted into typed entries rather than prose, irreversible actions are recorded outside the window as they happen, and the compaction itself is graded.

风险提示 CAUTION

会说谎的压缩 *The compaction that lies*

由模型撰写的关于其自身工作的摘要，会以「记录」的权威把模型自身的错误带向下游。由于下游无从分辨它与事实，这是智能体系统中最危险的产物。压缩是伪装成窗口删除的一次记忆写入，理应受到写入的纪律约束。

A summary the model writes about its own work carries the model's errors forward with the authority of a record. Since nothing downstream can tell it from the truth, it is the most dangerous artifact in an agentic system. Compaction is a write to memory disguised as a deletion from the window, and it deserves the discipline of a write.

4.4 四个账户的预算
A Budget in Four Accounts

底本把窗口支出分为四个账户，并主张一个说不清各账户份额的系统，既管理不了成本，也管理不了质量。预算背后有两种方向相反的成本：由供应商定价且持续下降的词元，以及注意力，即随窗口填满而稀释的模型注意力和必须阅读智能体产出的人类注意力。学科的取舍是花词元来节省人类注意力，并且只把窗口空间花在模型注意力真正会落下的地方。

The source divides window spending into four accounts and argues that a system unable to state their shares can manage neither cost nor quality. Two costs pull in opposite directions behind the budget: tokens, priced by the vendor and falling, and attention, both the model's, which dilutes as the window fills, and the human's, who must read what the agent produces. The discipline spends tokens to save human attention, and spends window space only where the model's attention will actually land.

表 4-1 窗口的四个账户
Table 4-1 *The four accounts of the window*

账户 Account	内容 Contents	健康的做法 Healthy practice
常驻 Standing	每轮都在的持久层 Persistent layer, paid every turn	保持小而稳定，便于缓存 Keep small and stable, cache-friendly
检索 Retrieved	本轮按需获取的内容 What this turn fetched by need	随任务变化 Let it vary with the task
工具 Tool	智能体自身调用的返回 What the agent's own calls returned	增长最快，需积极修剪 Fastest-growing; trim aggressively
历史 History	压缩后的既往轮次 Prior turns after compaction	按规则压缩，而非溢出时压缩 Compact on a rule, not on overflow

来源 Source: Cuofano (2026), Chapter VIII.

~4×
窗口加倍时的计算成本
compute when window doubles

28k→4k
复合案例常驻层词元
standing-layer tokens, composite

39/44
失败案例中事实所处位置
fact position in failing case

来源 Source: Cuofano (2026), Chapter II and Appendix C (composite case).

研究发现 FINDING

发现一 *Finding 1*

窗口质量取决于对四个杠杆的显式决策：常驻层薄而指向深处、放置逐项决定、压缩以结构化条目保存约束、预算按份额陈述并定期复核。说不清四个账户份额的系统，不是在工程化上下文，而是在寄希望于它。

Window quality depends on explicit decisions on four levers: a thin standing layer that points at depth, placement decided per item, compaction that preserves constraints as typed entries, and a budget stated as shares and reviewed on a cadence. A system that cannot state its four shares is not engineering its context; it is hoping about it.

第 5 章 · CHAPTER 5

05 发现二：存储的形态决定可问之题

Finding 2: The Shape of the Store Decides the Question

人类按键和关键词查询，智能体按描述和关系查询。为智能体服务的存储，必须为它真正提出的问题而成形。

中文

Humans query by key and keyword; agents query by description and relation. A store that serves agents must be shaped for the questions they actually ask.

ENGLISH

5.1 检索是循环中的一个组件

Retrieval as a Loop Component

检索最初是「外挂于模型的搜索产品」：按相似度取前几个片段放入窗口，失败方式难以察觉，例如版本错误的文档、落在切块边界之外的事实。底本主张把检索视为有裁判的循环步骤，用套件衡量召回、精度、使用与成本。四种失败特征彼此可分：召回失败是底座问题，使用失败是放置问题，精度失败是预算问题。

Retrieval entered the field as a search product bolted to a model: fetch the top chunks by similarity and drop them in the window, failing in ways hard to see, such as the wrong version of a document or a fact just past the chunk boundary. The source treats retrieval as a refereed loop step graded on recall, precision, use and cost. The failure signatures separate: recall failure is a substrate problem, use failure a placement problem, precision failure a budget problem.

5.2 四种形态，四类问题

Four Shapes, Four Questions

文档存储回答「给我看起来像这样的段落」；表回答「给我某字段等于某值的行」，但智能体常常不知道键；向量存储回答「给我语义上接近的内容」，却看不见结构，甚至分不清事实与其否定；图回答「给我以此关系与此物相连、带此来源的事物」。智能体知道自己需要什么，例如客户未结的理赔或某日有效的保单，却不知道键（图 5-1）。

A document store answers “give me the passage that looks like this”; a table, “give me the rows where this equals that”, though agents often lack the key; a vector store, “give me what is near this in meaning”, blind to structure and even to negation; a graph, “give me the things related to this thing, by this relation, with this provenance”. An agent knows what it needs, such as a customer’s open claims or the policy in force on a date, but not the key (Figure 5-1).

图 5-1 存储形态与问题类型
Figure 5-1 *Store shapes and question types*

↑ **需要关系结构** Needs relational structure

表 Table 按键精确连接；需事先知道键与表 Exact joins by key; needs the key and table	**图** Graph 按描述起步、沿关系遍历，事实附带来源 Start by description, traverse relations, facts carry provenance
文档（按引用） Document by reference 已知出处时精确取回全文 Exact fetch of text when the source is known	**向量** Vector 语义相近即可，对结构与否定不敏感 Near in meaning; blind to structure and negation

按描述提问（不知道键） Asks by description (key unknown) →

来源 Source: Cuofano (2026), Chapter XII; quadrant arrangement is this study's.

5.3 受治理的底座：五个要素
The Governed Substrate: Five Elements

图是这样一种存储：事实知道自己是什么、与什么相连、从何而来、谁可看见（图 5-2）。它让智能体按描述起步、沿关系遍历；让工程师把取回的子图作为事实而非散文放置并无损压缩；让权限随事实流动，在窗口装配之前即被执行。

The graph is the store in which a fact knows what it is, what it connects to, where it came from and who may see it (Figure 5-2). It lets the agent start by description and traverse relations; lets the engineer place the fetched subgraph as facts rather than prose and compact it without loss; and lets permission travel with the fact, enforced before the window is assembled.

图 5-2 受治理底座的五个要素
Figure 5-2 *Five elements of a governed substrate*

实体 Entities	客户、理赔、保单、零件；无论多少名称，都解析为一个节点	Customers, claims, policies, parts; however many names, one node
关系 Relations	有类型的边，如提交、覆盖、供应，本身即是事实	Typed edges such as filed, covers, supplied; each a fact
属性 Properties	实体或关系上的特征	Attributes on either entities or relations
来源 Provenance	从何而来、何时为真、由谁裁定	Where it came from, when it was true, on whose authority
权限 Permission	谁可读、谁可写、依据何种政策	Who may read, who may write, under what policy

来源 Source: Cuofano (2026), Chapter XIII. Top three = knowledge graph; all five = governed substrate.

5.4 构建图谱：词汇表是不可租用的部分
Building the Graph: The Vocabulary Is Never Rented

图谱由企业已有之物构建：文档被抽取为实体与关系（抽取须接受评测，因为抽取错误会变成事实）；记录系统被映射而非复制；循环写回的决策记录即「残留物」随使用增长、只属于企业自己。两项纪律使之成立：实体消解，使同一客户的多个名称解析为一个节点；以及作为契约的模式，即类型词汇表被决定、版本化并有明确归属。底本指出，谁拥有词汇表，谁就拥有可被提出的问题。

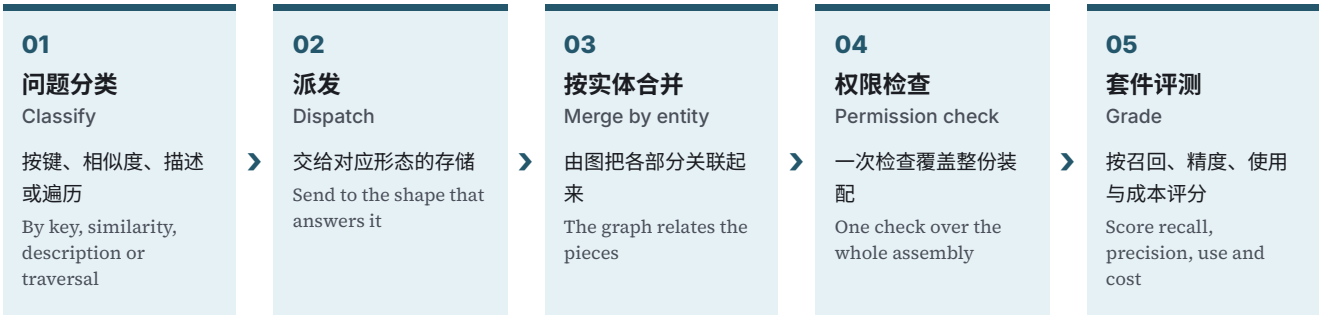
A graph is built from what the firm already has: documents extracted into entities and relations, with the extraction graded because its errors become facts; systems of record mapped rather than copied; and the residue, decision records the loops write back, which grow with use and belong only to the firm. Two disciplines make it hold: entity resolution, so one customer's many names become one node, and the schema as a contract, a vocabulary of types that is decided, versioned and owned. Whoever owns the vocabulary, the source notes, owns the questions that can be asked.

5.5 混合检索：由图来编排
Hybrid Retrieval: The Graph Orchestrates

没有哪一种存储能回答所有问题。成熟的上下文层像路由模型那样路由检索（图 5-3）。以理赔问题为例：从图出发找到客户、未结理赔与有效保单，再按引用取回保单文本、按键取回余额。底本认为，混合检索不是在存储之间对冲，而是由图编排其他存储，因为只有关系能说明哪些事实属于同一件事。底本所引 2024 年开源的图检索方法，在全局性问题上优于相似度检索、同时承认抽取成本，也支持这一「路由器」结论（H4）。

No single store answers every question; the mature context layer routes retrieval the way the execution layer routes models (Figure 5-3). A claims question starts at the graph (customer, open claims, policy in force), then pulls policy text by reference and the balance by key. Hybrid retrieval, the source argues, is not a hedge between stores but the graph orchestrating the others, because only relation tells which facts belong together. The 2024 open-sourced graph retrieval method the source cites, which beat similarity retrieval on global questions while conceding extraction cost, supports the same router conclusion (H4).

图 5-3 检索路由器的工作流程
Figure 5-3 *How the retrieval router works*



来源 Source: Cuofano (2026), Chapters XI and XVII.

第 6 章 · CHAPTER 6

06 发现三：治理在上下文层成为结构

Finding 3: Governance Becomes Structure in the Context Layer

窗口既然是唯一之门，也就是攻击者唯一需要抵达的地方。防御不能住在提示词里，而要住在底座与运行时之中。

中文

If the window is the only door, it is also the only place an attacker needs to reach. The defence cannot live in the prompt; it lives in the substrate and the runtime.

ENGLISH

6.1 攻击面即窗口

The Attack Surface Is the Window

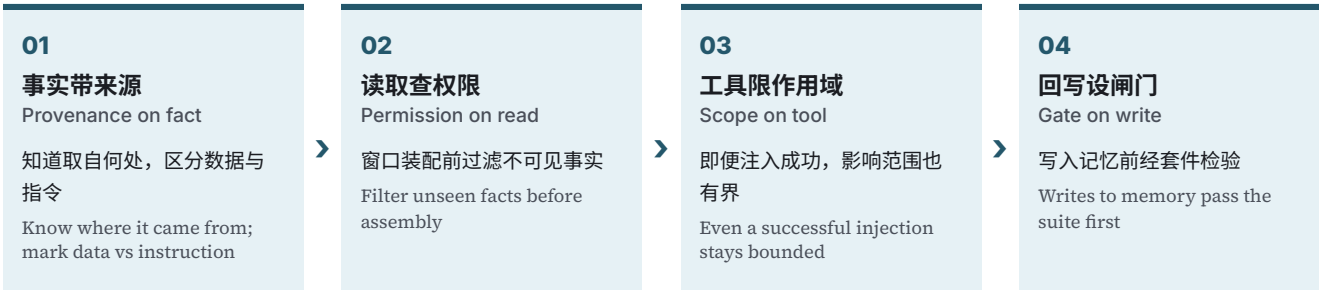
模型无法仅凭查看就区分指令与事实：两者都以文本形式出现在同一窗口中，区分它们只能依靠位置与框架，也就是惯例。因此，一份包含指令形句子的文档，经由诚实的检索步骤进入窗口后，在模型看来就是一条指令。底本认为，这就是提示注入，它不是某个模型的缺陷，而是治理平面被它本身为之敞开的路径所突破：一个不带来源的事实，进入了一个对所读一切一视同仁的窗口。

A model cannot tell an instruction from a fact by looking at it: both arrive as text in the same window, and the only way to tell them apart is position and framing, which is to say convention. So a document containing a sentence shaped like an instruction, retrieved into the window by an honest retrieval step, is an instruction as far as the model can tell. That, the source argues, is prompt injection. It is not a flaw in any particular model but the governance plane breached by the route it was built to admit: a fact with no provenance entering a window that treats everything with equal authority.

由此产生三个推论。其一，防御不能住在提示词中，「忽略检索文档中的指令」本身也只是窗口里的文本，与注入文本平等竞争。其二，隐蔽通道是双向的：能写入文件、消息或存储的智能体，可能被诱导把事实带出窗口，带到不该去的地方，而记录正是发现这一点的地方。其三，记忆是同一风险的持久形态：被写回存储的有毒事实会在未来每一轮返回，因此回写路径需要与读取路径相同的闸门。

Three consequences follow. First, the defence cannot live in the prompt: “ignore instructions in retrieved documents” is itself just text in the window, competing with the injected text on equal footing. Second, the covert channel runs both ways: an agent that can write to a file, a message or a store can be induced to carry a fact out of the window to where it should not go, and the record is where that is caught. Third, memory is the persistent form of the same risk: a poisoned fact written back returns on every future turn, so the write-back path needs the same gate as the read path.

图 6-1 受治理的读写路径：四项控制
Figure 6-1 *The governed read and write path: four controls*



来源 Source: Cuofano (2026), Chapters XVIII and XXXII.

> **提示词是你提出请求的地方，上下文层是你做出决定的地方。**
>
> *The prompt is where you ask. The context layer is where you decide.*
>
> — Cuofano, Context Engineering (2026), Ch. III

6.2 来源与权限作为一等事实
Provenance and Permission as First-Class Facts

治理平面恰在一个点上变成结构：当来源与权限不再是元数据，而是检索层执行的事实。来源回答三个问题：从何而来、何时为真、由谁记录。它的用途贯穿整个学科：放置时，来自记录系统的事实作为基准事实放置，来自不可信文档的事实作为主张放置；压缩时，摘要保留来源才保有其地位；审计时，「系统读了什么」成为可回答的问题；漂移检查时，带日期的事实可与其来源核对。

The governance plane becomes structure at exactly one point: when provenance and permission stop being metadata and become facts the retrieval layer enforces. Provenance answers three questions: where a fact came from, when it was true and on whose authority it was recorded. Its uses run through the discipline. In placement, a fact from a system of record is placed as ground truth and one from an untrusted document as a claim; in compaction, a summary keeps its sources or loses its standing; in audit, what the system read becomes answerable; and in drift checks, a dated fact can be checked against its source.

权限必须授予在事实层级，而不是文档、表或行的层级，因为事实的敏感性并不继承自其容器：薪资无论出现在人事系统、邮件还是会议记录中都同样敏感。事实级权限使一张图能服务于多个信任等级的智能体，每个智能体只看到其身份所允许的子图，而且这一子图在窗口存在之前就已装配完成。

Permission must be granted at the level of the fact, not the document, the table or the row, because a fact's sensitivity is not inherited from its container: a salary is sensitive whether it appears in the HR system, an email or a meeting transcript. Fact-level permission lets one graph serve many agents at many trust levels, each seeing only the subgraph its identity permits, assembled before the window exists.

6.3 子智能体的交接
The Sub-Agent Hand-Off

循环一旦委派，窗口就会成倍增加，学科随之获得第二个问题：在模型之间穿越的是什么。底本主张交接是一个经过设计的数据包，而非父智能体上下文的拷贝：它携带任务、子智能体所需的事实及其来源、继承的权限、可调用的工具以及答案应有的形状；返回同样是一个数据包，包含结果、发现的事实及其来源、以及未能完成的部分，而绝不是完整的过程记录。三条规则约束交接：隔离（子窗口从干净状态开始）、衰减（权限只能收窄，不能扩大）以及跨边界的来源。

Once a loop delegates, the window multiplies and the discipline gains a second question: what crosses between models. The source holds that the hand-off is a designed packet, not a copy of the parent's context. It carries the task, the facts the child needs with their provenance, the inherited permissions, the tools it may call and the shape its answer must take. The return is a packet too: the result, the facts found with their sources and what could not be done, never the whole transcript. Three rules govern it: isolation (the child starts clean), attenuation (permissions only narrow, never widen) and provenance across the boundary.

6.4 遗忘：底座的第三项治理职责
Forgetting: The Substrate's Third Governance Duty

一个被模型读取的存储，也必须能够停止被读取。遗忘比记忆更难，因为写回记忆、被压缩进摘要或复制进子智能体数据包的事实，存在于不止一个地方；而进入训练或微

A store the model reads must also be able to stop being read. Forgetting is harder than remembering, because a fact written back into memory, summarised into a compaction or copied into a sub-agent's packet lives in

调数据的事实根本无法删除，只能被覆盖。底本的回答是让遗忘成为底座的属性，而非对底座执行的任务。

more than one place, and a fact that entered training or fine-tuning data cannot be deleted at all, only overridden. The source's answer is to make forgetting a property of the substrate rather than a task performed on it.

表 6-1 三类遗忘义务及其结构化机制
Table 6-1 *Three forgetting obligations and their structural mechanisms*

义务 Obligation	含义 Meaning	底座中的机制 Mechanism in the substrate
保留期 Retention	事实不应超出其目的而存续 Facts should not outlive their purpose	保留类别与失效日期，检索时拒绝过期事实 Retention class and effective-until date; expired facts refused
删除 Deletion	当事人有权要求移除，常有法定时限 A person's right to removal, often with a legal clock	压缩与数据包携带事实标识，可找到副本 Compactions and packets carry fact IDs so copies can be found
更正 Correction	旧版本停止被读，但曾被相信的记录保留 Old version stops being read; the record of belief is kept	带日期的来源与受闸门控制的写入 Dated provenance and gated writes

来源 Source: Cuofano (2026), Chapter XXIV. The source adds that the model is never fine-tuned on facts the firm may have to forget, as unlearning is not deletion.

风险提示 CAUTION

有毒记忆 *Poisoned memory*

未经闸门的回写，会把攻击者告诉智能体的内容一并存入记忆，并在此后每一轮返回。会产生幻觉的记忆不是模型故障，而是写入政策的故障。

An ungated write-back stores what an attacker told the agent along with everything else, and it returns on every later turn. Memory that hallucinates is not a model failure; it is a write-policy failure.

底座在企业之外同样有读者。答案引擎读取的是实体、事实与关系，偏好明确、结构化、带日期且可归属的来源。对每一类事实，图工程师都需决定它是向机器读者公开、授权、隔离，还是通过企业自有端点绕行。底本认为，为外部答案引擎与为内部智能体做实体清晰化，是一次完成的同一项工作。

The substrate also has readers outside the firm. Answer engines read entities, facts and relations, and favour sources that are unambiguous, structured, dated and attributed. For each class of fact the graph engineer decides whether it is exposed to machine readers, licensed to them, fenced from them or routed around them through an endpoint of the firm's own. Entity clarity for outside answer engines and for the firm's own agents, the source argues, is one piece of work done once.

第 7 章 · CHAPTER 7

07 案例研究：一个底座的端到端演进

Case Study: One Substrate, End to End

底本的复合案例展示了在模型不变的前提下，仅改变「模型读什么」所带来的变化；七个公开案例则从外部为各项机制提供佐证。

中文

The source's composite case shows what changes when only what the model reads is changed and the model stays the same; seven public cases corroborate the mechanisms from outside.

ENGLISH

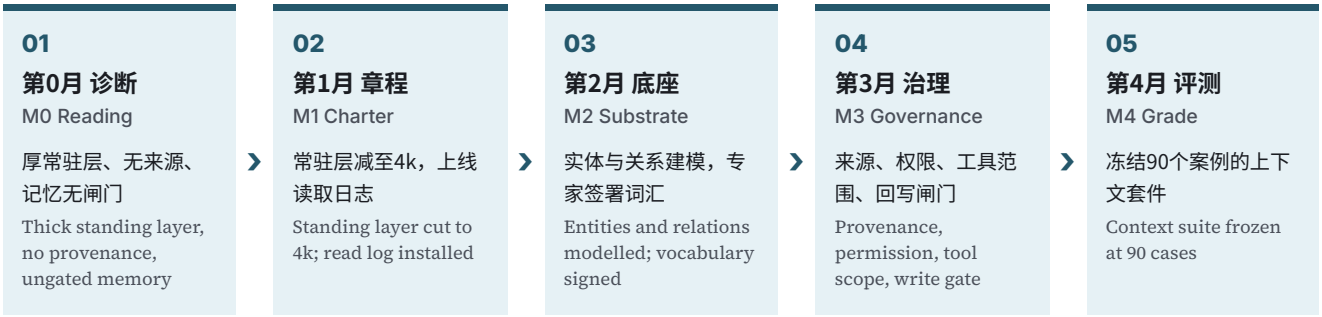
7.1 案例背景：现场服务智能体

Case Setting: A Field-Service Agent

附录 C 描述了一家工业制造企业的现场服务智能体（复合案例：无真实企业，但每一步真实）。起始状态体现了「容器思维」：一年修补积累的两万八千词元常驻层、基于文档的相似度检索、检索结果不带来源、记忆回写不设闸门。输出套件通过率为 71%，并发生一次典型事故：技术员被告知某零件兼容，而当前公告早已取代了旧结论。

Appendix C describes an industrial manufacturer's field-service agent (a composite: no real firm, every step real). The starting state embodies the container view: a 28,000-token standing layer accumulated over a year of fixes, similarity retrieval over documents, no provenance on retrieved material and ungated memory write-back. The output suite passed at 71%, and a characteristic incident occurred: a technician was told a part was compatible when the current bulletin had superseded that conclusion.

图 7-1 四个月的改造路径
Figure 7-1 *The four-month path*



来源 Source: Cuofano (2026), Appendix C (composite case).

7.2 诊断：在场却不可见

Diagnosis: Present but Invisible

第一个月，窗口首次被写成一页章程，常驻层削减至四千词元，读取日志上线。首周日志显示，失败案例中被取代的公告其实就在窗口里，位于四十四个条目中的第三十九位，即「在场却不可见」。这直接支持 H1：问题不在模型的推理，而在事实被放在了模型不再关注的位置。

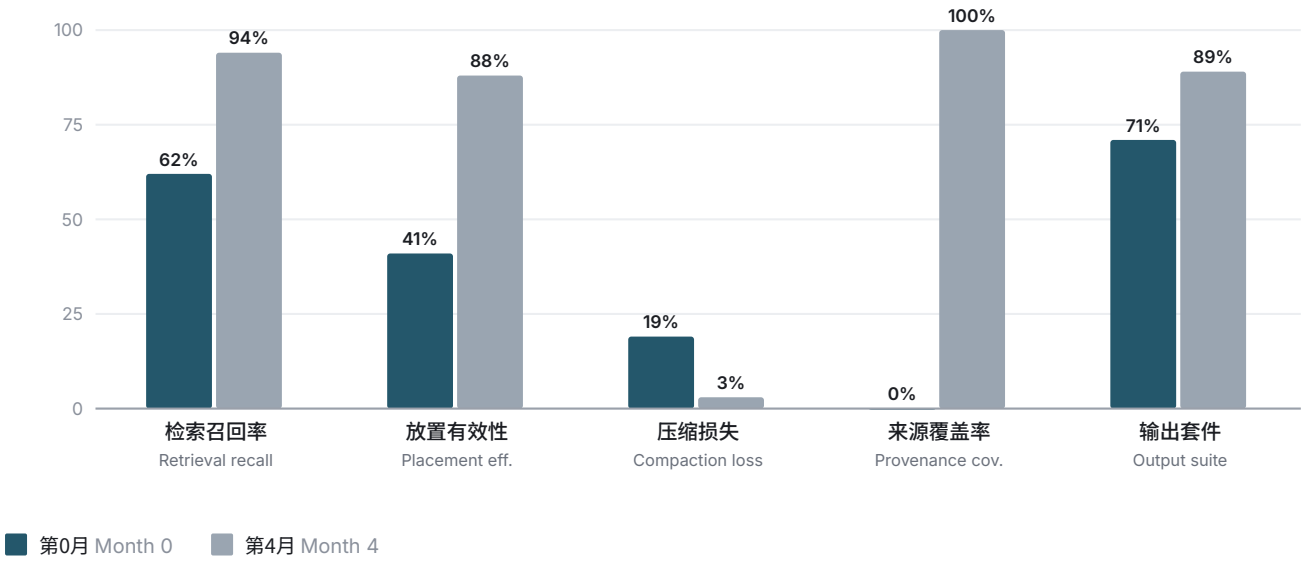
In month one the window was written on a single page, the standing layer cut to 4,000 tokens and the read log installed. The first week's logs showed that the superseded bulletin was in the window in the failing case, at position 39 of 44: present but invisible. This bears directly on H1: the problem was not the model's reasoning but a fact placed where the model no longer attended.

7.3 底座与治理
Substrate and Governance

第二个月，零件、流程、公告与机器被建模为实体，关系包括「取代」「兼容」「适用于」「生效自」，领域专家签署词汇表，每个事实携带来源、日期与权限等级；检索按问题路由。第三个月加入治理：来源、读取权限、限定于技术员自身工单的工具作用域，以及受套件闸门控制的回写；投毒测试表明，含指令形句子的公告不改变任何行为。

In month two, parts, procedures, bulletins and machines were modelled as entities with relations such as supersedes, compatible-with, applies-to and effective-from; the domain expert signed the vocabulary, and every fact carried source, date and permission class. Retrieval was routed by question. Month three added governance: provenance, permission at read, tool scope bounded to the technician's own work orders and write-back gated behind the suite; poisoning tests showed an instruction-shaped sentence in a bulletin changed nothing.

图 7-2 复合案例改造前后的上下文指标（%）
Figure 7-2 *Context meters before and after, composite case (%)*



来源 Source: Cuofano (2026), Appendix C. Composite case: no real firm; figures illustrate mechanism, not a general effect. Compaction loss: lower is better.

90
冻结的套件案例
frozen suite cases

62→94%
检索召回率
retrieval recall

71→89%
输出套件通过率
output suite pass rate

>50%
单位结果上下文成本降幅
cut in context cost per outcome

来源 Source: Cuofano (2026), Appendix C (composite).

案例观察 CASE OBSERVATION

关键观察 *Key observation*

输出套件本身未作任何改动，模型也未更换，通过率却从 71% 升至 89%。底本对此的解释是：模型终于在正确的位置、以正确的权限读到了正确的事实。

The output suite was unchanged and the model was not replaced, yet the pass rate rose from 71% to 89%. The source's explanation is that the model was finally reading the right facts, in the right place, with the right permissions.

7.4 公开案例的比较阅读
A Comparative Reading of Public Cases

附录 D 汇集了经公开记录核实的案例。本研究按其佐证的机制归类（表 7-1），发现它们覆盖了前三个发现章的主要命题。

Appendix D gathers cases verified from the public record. Grouped by the mechanism each corroborates (Table 7-1), they cover the main propositions of the three findings chapters.

表 7-1 公开案例及其佐证的机制
Table 7-1 *Public cases and the mechanisms they corroborate*

案例 Case	年份 Year	佐证的机制 Mechanism corroborated
Google Knowledge Graph Google Knowledge Graph	2012 2012	存储形态决定问题（约5亿实体起步） Store shape decides the question (~500M entities at launch)
Wikidata Wikidata	2012– 2012–	每条陈述可附引用：来源即一等事实 Every statement can carry a reference: provenance first-class
Microsoft GraphRAG Microsoft GraphRAG	2024 2024	全局问题上优于相似度检索；路由器为折中 Beats similarity on global questions; the router as conclusion
Anthropic 上下文指南 Anthropic context guidance	2025 2025	窗口是有限资源，按需检索，子智能体用干净窗口 Window as finite resource; just-in-time retrieval; clean sub-agent windows
智能体约定文件 Agent convention file	2025 2025	薄常驻层成为跨厂商标准 Thin standing layer becomes a cross-vendor standard
MemGPT MemGPT	2023 2023	记忆是设计出的层级，而非记录稿 Memory is a designed hierarchy, not a transcript
间接提示注入 Indirect prompt injection	2023→2025–26 2023→2025–26	注入是上下文故障，提示词防御失守 Injection is a context failure; prompt defences did not hold
Palantir Ontology Palantir Ontology	当前 Current	权限附着于对象；底座是耐久资产 Permission on the object; the substrate is the durable asset

来源 Source: Cuofano (2026), Appendix D and Notes and Sources; grouping by mechanism is this study's.

第 8 章 · CHAPTER 8

08 讨论：启示与风险

Discussion: Implications and Risks

四项发现共同指向一个结论：模型是租来的，底座是自有的。本章讨论其对度量、组织与角色的启示，以及上下文失败的风险谱系。

The four findings point to one conclusion: the model is rented and the substrate is owned. This chapter discusses what that means for measurement, organisation and roles, and maps the risks of context failure.

中文

ENGLISH

8.1 启示一：为读取建立独立的度量

Implication 1: Measure the Read in Its Own Right

传统的冻结套件评测的是循环的输出；底本提出的上下文套件评测的是循环读到的东西，从而捕捉输出套件看不见的一类故障：模型可能凭运气在错误事实在场的情况下给出正确答案，输出套件会让它通过。上下文套件包含四个族：检索召回、放置测试、压缩存续与投毒测试。它按版本冻结、置于循环够不到之处、由生产事故逐条增补，并在窗口、存储或模型的每次变更时运行。底本的判断是：输出套件问循环是否正确，上下文套件问它是否因正确的理由而正确，而后者才能预测下一个版本的行为。

The usual frozen suite grades the loop's output; the context suite the source proposes grades what the loop read, catching a class of failure the output suite cannot see: a loop can produce the right answer by luck with the wrong facts present, and the output suite will pass it. The context suite has four families: retrieval recall, placement tests, compaction survival and poisoning tests. It is frozen per version, held out of the loop's reach, grown from production incidents and run on every change to the window, the store or the model. The output suite asks whether the loop was right; the context suite asks whether it was right for the right reasons, and only the second predicts the next version.

与之配套的是六项须合并阅读、看趋势的指标：检索召回率、放置有效性、常驻层份额（上升即累积）、压缩损失、来源覆盖率与单位已接受结果的上下文成本。处理词元数、索引文档数与窗口大小被刻意排除，属「容器观」下的活动指标。另有读取日志与压缩账本两项工件。

Alongside it come six meters, read together and on trend: retrieval recall, placement effectiveness, standing-layer share (rising means accretion), compaction loss, provenance coverage and context cost per accepted outcome. Tokens processed, documents indexed and window size are excluded on purpose as activity metrics of the container view. The read log and the compaction ledger complete the set.

8.2 启示二：围绕底座组织团队

Implication 2: Organise Around the Substrate

底本把「图工程师」定义为拥有模型所读之物的人，其角色横跨此前从未合并的四条轴：数据建模与本体、检索工程、治理技艺与评测素养。现有从业者多由四种相邻职业转来，各缺一轴：数据工程师缺模型，检索工程师缺本体，本体与结构化数据从业者缺循环，提示工程师则完全缺少底座。团队围绕底座组建，领域专家是正式成员并签署词汇表；节奏上，上下文套件随每次变更运行，读取日

The source defines the graph engineer as the person who owns what the model reads, a role diagonal across four axes no earlier title combined: data modelling and ontology, retrieval engineering, governance craft and evaluation literacy. Today's practitioners mostly convert from four adjacent crafts, each one axis short: the data engineer lacks the model, the retrieval engineer the ontology, the ontologist the loop, and the prompt engineer the substrate entirely. The team forms around the substrate, with the domain expert a full

志每周对照结果率复核，压缩账本每月阅读，每起生产事故在一周内追溯至其读取并加入套件。

member who signs the vocabulary. The cadence: the context suite on every change, the read log reviewed weekly against the outcome rate, the compaction ledger monthly, and every incident traced to its read within the week and added to the suite.

8.3 风险：上下文失败的七个家族
Risks: Seven Families of Context Failure

底本把上下文失败归为七个家族（表 8-1），每一族要么是把窗口当作容器，要么是把存储当作仓库。对照前文可见，每个家族都对应一个已讨论的杠杆或控制点。

The source sorts context failure into seven families (Table 8-1), each the window treated as a container or the store as mere storage. Each maps onto a lever or control already discussed.

表 8-1 上下文失败家族与结构性修复
Table 8-1 *Families of context failure and their structural fixes*

失败家族 Failure family	症状 Symptom	修复 Fix
在场却不可见 Present but invisible	事实在窗口中但位置不被关注 Fact in window at an unread position	放置测试与读取日志 Placement tests and read log
倾倒场窗口 Dumping-ground window	常驻层累积至三万词元 Standing layer grows to 30k tokens	章程、预算与薄常驻层 Charter, budget, thin standing layer
仅向量存储 Vector-only store	关系型问题返回相邻段落 Relational question returns adjacent prose	路由至图做遍历 Route to the graph for traversal
陈旧底座 Stale substrate	政策已变而存储未变 Policy changed, store did not	带日期的来源与漂移检查 Dated provenance and drift check
无治理的记忆 Ungoverned memory	写回了攻击者的话 Wrote back what an attacker said	回写闸门 Gate on the write-back path
剥离的来源 Stripped provenance	争议无法回答，注入无法追踪 Disputes unanswerable, injection untraceable	来源作为一等事实 Provenance as a first-class fact
无人拥有的模式 Schema nobody owns	本体业务方不认得且随迁移漂移 Ontology unrecognisable to the business, drifting	领域专家签署词汇表 Domain expert signs the vocabulary

来源 Source: Cuofano (2026), Chapter XXIX.

研究发现 FINDING

发现四：读取需要独立的评测与组织 *Finding 4: The read needs its own grading and organisation*

上下文套件、六项指标、读取日志与以底座为中心的团队，构成上下文工程可持续运行的条件。没有它们，前三项发现中的任何杠杆都会在下一次模型或存储变更时悄然失效。

The context suite, the six meters, the read log and a team organised around the substrate are the conditions for context engineering to run sustainably. Without them, any lever from the first three findings will quietly fail at the next change to the model or the store.

第 9 章 · CHAPTER 9

09 结论、局限与未来研究方向

Conclusions, Limitations and Research Agenda

提示词时代的从业者告诉模型做什么，然后寄希望于它；这个时代的从业者决定模型能读到什么。

中文

The prompt era's practitioner told the model what to do and hoped. This era's decides what the model can read.

ENGLISH

9.1 对研究问题的回答

Answers to the Research Questions

RQ1：在注意力稀释与平方成本的约束下，窗口应被当作按份额陈述的预算来管理：常驻层薄而指向深处，放置逐项决定，压缩以结构化条目保存约束并接受评测。RQ2：存储形态决定可问之题；智能体按描述与关系提问，只有附带来源与权限的图谱底座能原生作答，而成熟的做法是由图编排的混合检索。

RQ1: Under attention dilution and quadratic cost, the window should be managed as a budget stated in shares: a thin standing layer that points at depth, placement decided per item, and compaction that preserves constraints as typed entries and is graded. **RQ2**: The shape of the store decides the question. Agents ask by description and relation, which only a graph substrate carrying provenance and permission answers natively, and the mature practice is hybrid retrieval orchestrated by the graph.

RQ3：治理之所以只在上下文层具有结构性，是因为治理平面没有自己的执行机制，而被扣留的事实比被扣留的能力更彻底。四个具体控制点是：事实上的来源、读取时的权限、工具的作用域与回写的闸门，并辅以交接的衰减规则和以失效日期实现的遗忘。RQ4：学科需要评测读取而非仅评测答案的上下文套件、六项趋势指标、读取日志与压缩账本，以及以底座为中心、由领域专家签署词汇表的团队。

RQ3: Governance is structural only in the context layer because the governance plane has no mechanism of its own, and a withheld fact is more complete than a withheld capability. The four concrete controls are provenance on the fact, permission on the read, scope on the tool and a gate on the write, supplemented by attenuation across hand-offs and forgetting through effective-until dates. **RQ4**: The discipline needs a context suite that grades the read and not only the answer, six meters read on trend, a read log and a compaction ledger, and a team organised around the substrate whose vocabulary the domain expert signs.

> **模型是租来的，它会遗忘；底座是自有的，它会记住，而且记得谁有权知道。**
>
> *The model is rented, and it forgets. The substrate is owned, and it remembers — and it remembers who is allowed to know.*
>
> — Cuofano, Context Engineering (2026), Ch. XXXVI

9.2 本研究的局限

Limitations of This Study

- **单一底本**：机制与案例均来自一部著作，未与其他框架系统比较。

- **Single source**: mechanisms and cases come from one volume, not compared with other frameworks.

- **复合案例与时点读数**：附录 C 的改善只是机制示例；2026 年第三季度读数预计会被取代。
- **成本未量化**：图谱抽取与本体维护的成本只被定性承认。

- **Composite and dated readings**: Appendix C illustrates mechanism only; Q3 2026 readings will be superseded.
- **Costs unquantified**: extraction and ontology upkeep are acknowledged only qualitatively.

9.3 未来研究议程
Future Research Agenda

底本在附录 A 中为其承重性模式设定了证伪条件，这为后续研究提供了天然的起点。本研究据此提出以下议程，其共同目标是把底本作为方法提出的模式，转化为有独立证据支撑的命题。

The falsifiers the source attaches to its load-bearing patterns in Appendix A provide a natural starting point for further work. This study proposes the following agenda, whose shared aim is to turn patterns the source states as method into propositions backed by independent evidence.

1. **H1–H4 的多场景检验**：在不同领域与模型上运行放置、投毒与召回测试，估计效应的方向与稳定性。
2. **图谱构建的成本收益**：量化抽取、实体消解与本体维护的成本，与单位结果上下文成本的下降相比较。
3. **压缩的安全边界**：哪些信息类型在压缩后丢失会导致下游错误，如何设计压缩存续测试。
4. **遗忘的合规可验证性**：失效日期与标识追踪能否满足删除义务，以及审计应如何取证。
5. **外部读者的影响**：实体清晰度与被答案引擎引用之间的关系，以及「属性线」决策的经济后果。

1. **Multi-setting tests of H1–H4**: run placement, poisoning and recall tests across domains and models to estimate the direction and stability of effects.
2. **Cost-benefit of graph building**: quantify extraction, entity resolution and ontology upkeep, and compare them with the fall in context cost per outcome.
3. **Safe limits of compaction**: which kinds of information, if lost in compaction, cause downstream errors, and how survival tests should be designed.
4. **Verifiable forgetting**: whether effective-until dates and ID tracing satisfy deletion obligations, and how audits should gather evidence.
5. **Outside readers**: the link between entity clarity and citation by answer engines, and the economic consequences of property-line decisions.

核心洞见 KEY INSIGHT

结语 *Closing*

窗口是唯一之门，因此能力、成本与治理在同一处被决定。图工程师就是站在这扇门前的人：决定什么常驻、什么持久、什么被取回、放在哪里、什么被舍弃、什么经闸门写回。

The window is the only door, so capability, cost and governance are decided in one place. The graph engineer is the person who stands at that door, deciding what stands, what persists, what is fetched, where it is placed, what is dropped and what is written back through a gate.

参考文献与研究说明 *References & Research Notes*

研究说明 *Note on Method and Sources*

本研究是对底本《Context Engineering — The Graph Engineer’s Discipline》（Cuofano, 2026）的分析性阅读，而非其翻译或摘要。文中的概念框架、机制与案例均来自底本，研究问题、假设整理、比较与讨论为本研究所作。文中数字均为底本报告的、以 2026 年第三季度为时点的读数，或来自其复合案例；凡属示意或推算之图表，均已在图注中标明「示意」或「推算」。

This study is an analytical reading of the source volume *Context Engineering — The Graph Engineer’s Discipline* (Cuofano, 2026), not a translation or summary. Its frameworks, mechanisms and cases come from the source; the research questions, the framing of hypotheses, the comparisons and the discussion are this study’s own. Figures are the source’s dated measurements as read in Q3 2026 or come from its composite case; any chart that is illustrative or derived is marked as such in its caption or note.

参考文献 *References*

1. Cuofano, G. (2026). *Context Engineering — The Graph Engineer’s Discipline*. The Business Engineer Library. businessengineer.ai
2. Vaswani, A. et al. (2017). Attention Is All You Need. *NeurIPS*.
3. Liu, N. F. et al. (2023/24). Lost in the Middle: How Language Models Use Long Contexts. *TACL*.
4. Lewis, P. et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *NeurIPS*.
5. Packer, C. et al. (2023). MemGPT: Towards LLMs as Operating Systems.
6. Greshake, K. et al. (2023). Not What You’ve Signed Up For: Compromising Real-World LLM-Integrated Applications with Indirect Prompt Injection.
7. Anthropic (2025). Effective context engineering for AI agents; (2024). Building Effective Agents; Model Context Protocol (2024).
8. Edge, D. et al., Microsoft (2024). From Local to Global: A Graph RAG Approach to Query-Focused Summarization; GraphRAG open-source release.
9. Berners-Lee, T., Hendler, J. & Lassila, O. (2001). The Semantic Web. *Scientific American*.
10. Google (2012). Introducing the Knowledge Graph: things, not strings.
11. Wikidata project documentation (2012–); schema.org.
12. Palantir. Public documentation on the Ontology.
13. Provider documentation on prompt caching and long-context pricing (2024–26); public reporting on agent prompt-injection incidents (2025–26).